

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 4: Quadratic Equations

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Word problems

1. A right-angled triangle has a perimeter of 60 cm and a hypotenuse of 25 cm. Find the lengths of the other two sides. **CBSE 2025, 5 marks**

SOLUTION

Let one of the other sides be x cm. The two sides add up to $60 - 25 = 35$ cm, so the other side is $(35 - x)$ cm.

By the Pythagoras theorem, $x^2 + (35 - x)^2 = 25^2$.

$x^2 + 1225 - 70x + x^2 = 625$, so $2x^2 - 70x + 600 = 0$, that is $x^2 - 35x + 300 = 0$.

$(x - 15)(x - 20) = 0$, so $x = 15$ or $x = 20$.

If $x = 15$ the other side is 20, and if $x = 20$ the other side is 15.

Answer: 15 cm and 20 cm

2. A train covers 480 km at a uniform speed. Had it travelled 8 km/h slower, the journey would have taken 3 hours longer. Find the speed of the train. **CBSE 2025, 5 marks**

SOLUTION

Let the speed be x km/h. The usual time is $\frac{480}{x}$ hours and the slower time is $\frac{480}{x - 8}$ hours.

$$\frac{480}{x - 8} - \frac{480}{x} = 3$$

$480x - 480(x - 8) = 3x(x - 8)$, so $3840 = 3x^2 - 24x$

$x^2 - 8x - 1280 = 0$, so $(x - 40)(x + 32) = 0$

$x = 40$ or $x = -32$. A speed cannot be negative.

Check: $480 \div 40 = 12$ hours and $480 \div 32 = 15$ hours, which is 3 hours more.

Answer: 40 km/h

- 3.** A rectangular floor is covered exactly by 200 square tiles. With tiles whose side is 1 unit longer, only 128 tiles are needed. Let the side of an original tile be x units. (i) Form a quadratic equation in x . (ii) Write it in standard form. (iii) Find x by factorisation, OR solve the equation by the quadratic formula. **CBSE 2024, 4 marks, case study**

SOLUTION

(i) The floor area is the same in both cases: $200x^2 = 128(x + 1)^2$

(ii) $200x^2 = 128x^2 + 256x + 128$, so $72x^2 - 256x - 128 = 0$. Dividing by 8: $9x^2 - 32x - 16 = 0$

(iii) By factorisation: $9x^2 - 36x + 4x - 16 = 0$, so $9x(x - 4) + 4(x - 4) = 0$ and $(9x + 4)(x - 4) = 0$

$x = 4$ or $x = -\frac{4}{9}$. A length cannot be negative, so $x = 4$.

OR by the formula: $x = \frac{32 \pm \sqrt{1024 + 576}}{18} = \frac{32 \pm 40}{18}$, which gives $x = 4$ or $x = -\frac{4}{9}$.

Answer: the side of a tile is 4 units.

2 Nature of roots

- 4.** The equation $ax^2 + bx + c = 0$, $a \neq 0$, has real and equal roots. Which relation holds? (a) $a = \frac{b^2}{c}$ (b) $b^2 = ac$ (c) $ac = \frac{b^2}{4}$ (d) $c = \frac{b^2}{a}$ **CBSE 2024, 1 mark**

SOLUTION

For real and equal roots, $D = b^2 - 4ac = 0$.

So $b^2 = 4ac$, that is $ac = \frac{b^2}{4}$.

Answer: (c)

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